

TicketOutTheDoor1.R

gcrone14

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```
#### Ticket out-the-door (1) - Answer ####
```

NOTE: The data to use is starbucks_items.csv. This file cannot be found directly, but its constituent data files can be found on this Kaggle page: <https://www.kaggle.com/datasets/starbucks/starbucks-menu>. What I did was import the data from online, and merge two data sets from the base files: starbucks-menu-nutrition-drinks.csv (drinks), and starbucks-menu-nutrition-food.csv (food). The columns are as follows:

- **item** = The name of the item
- **type** = Whether the item is a food (Food) or a drink (Drink)
- **calories** = Number of calories in the item

The other variables are other nutritional info, including fat (g), carb (g), fiber (g), protein (g), sodium (mg), and protein (g). We begin by loading our packages and importing the data. Next, we clean the data so it's easier to analyse, before generating descriptive statistics and a visualization.

```
# Loading packages
library(here)
```

```
## here() starts at /Users/gcrone14/Desktop/R Projects/PSYC3031-F26
```

```
library(psych)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.3      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x ggplot2::%+%( ) masks psych::%+%( )
## x ggplot2::alpha() masks psych::alpha()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

# Loading the data
star_items <- read.csv(here("Data/starbucks_items.csv")) |>
  # For simplicity of viewing
  as_tibble()

# Checking data
str(star_items)

## tibble [290 x 9] (S3: tbl_df/tbl/data.frame)
## $ item      : chr [1:290] "Cool Lime Starbucks Refreshers Beverage" "Ombré Pink Drink" "Pink Drink"
## $ type      : chr [1:290] "Drink" "Drink" "Drink" "Drink" ...
## $ calories  : int [1:290] 45 NA NA 80 60 NA NA NA 110 0 ...
## $ fat_g     : num [1:290] 0 NA NA 0 0 NA NA NA 0 0 ...
## $ carb_g    : int [1:290] 11 NA NA 18 14 NA NA NA 28 0 ...
## $ fiber_g   : int [1:290] 0 NA NA 1 1 NA NA NA 0 0 ...
## $ protein   : int [1:290] 0 NA NA 0 0 NA NA NA 0 0 ...
## $ sodium    : int [1:290] 10 NA NA 10 10 NA NA NA 5 0 ...
## $ protein_g : int [1:290] NA NA NA NA NA NA NA NA NA NA ...

```

```
glimpse(star_items)
```

```

## Rows: 290
## Columns: 9
## $ item      <chr> "Cool Lime Starbucks Refreshers Beverage", "Ombré Pink Drin~
## $ type      <chr> "Drink", "Drink", "Drink", "Drink", "Drink", "Drink", "Drink~
## $ calories  <int> 45, NA, NA, 80, 60, NA, NA, NA, 110, 0, NA, 130, 140, 130, N~
## $ fat_g     <dbl> 0.0, NA, NA, 0.0, 0.0, NA, NA, NA, 0.0, 0.0, NA, 2.5, 2.5, 2~
## $ carb_g    <int> 11, NA, NA, 18, 14, NA, NA, NA, 28, 0, NA, 21, 23, 21, NA, N~
## $ fiber_g   <int> 0, NA, NA, 1, 1, NA, NA, NA, 0, 0, NA, 0, 0, 0, NA, NA, NA, ~
## $ protein   <int> 0, NA, NA, 0, 0, NA, NA, NA, 0, 0, NA, 5, 5, 5, NA, NA, NA, ~
## $ sodium    <int> 10, NA, NA, 10, 10, NA, NA, NA, 5, 0, NA, 65, 90, 65, NA, NA~
## $ protein_g <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~

```

Here we do a simple data cleaning/modification. Note that this step is technically optional in this case. It is nice to clean the data up a tad.

```

star_items_final <- star_items |>
  # You may not wish to select only these variables,
  # but it's up to you.
  select(type, fat_g, carb_g) |>
  # We filter out missing data because it doesn't
  # provide any useful information.
  filter(!is.na(fat_g) & !is.na(carb_g))

# Descriptive stats for `type`
table(star_items_final$type)

```

```

##
## Drink  Food
##    92   113

```

The above shows that there are slightly more food items (113) than drink items (92).

(Note: You may show a bar chart of this result, but I find it superfluous since we only have two values.)

```
# Descriptive stats for fat and carbs
```

```
star_items |>
  select(fat_g, carb_g) |>
  psych::describe()
```

```
##          vars   n mean    sd median trimmed   mad min max range skew kurtosis
## fat_g      1 205 10.06  9.66     7   9.10 10.38    0  37   37 0.59   -0.92
## carb_g     2 205 33.97 17.61    35  34.02 17.79    0  80   80 0.01   -0.47
##              se
## fat_g    0.67
## carb_g   1.23
```

On average, items at Starbucks tend to contain more grams of carbohydrates ($M = 33.97$, $SD = 17.61$, $Mdn = 35$) than grams of fats ($M = 10.06$, $SD = 9.96$, $Mdn = 7.0$).

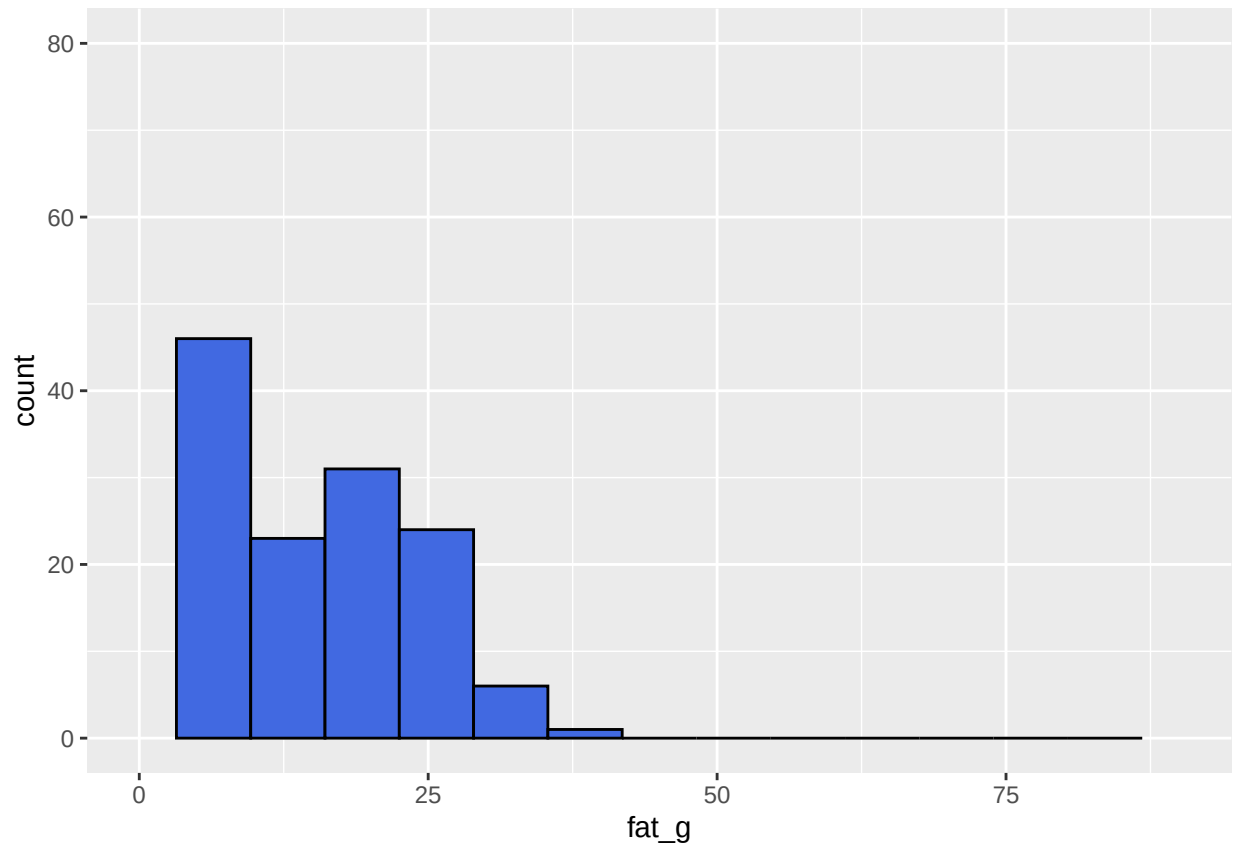
(Note: Reporting the median is usually optional; I did so because there was a large discrepancy between the mean and median for fats).

It's often helpful to visualize the distributions of continuous data

```
# For fats
```

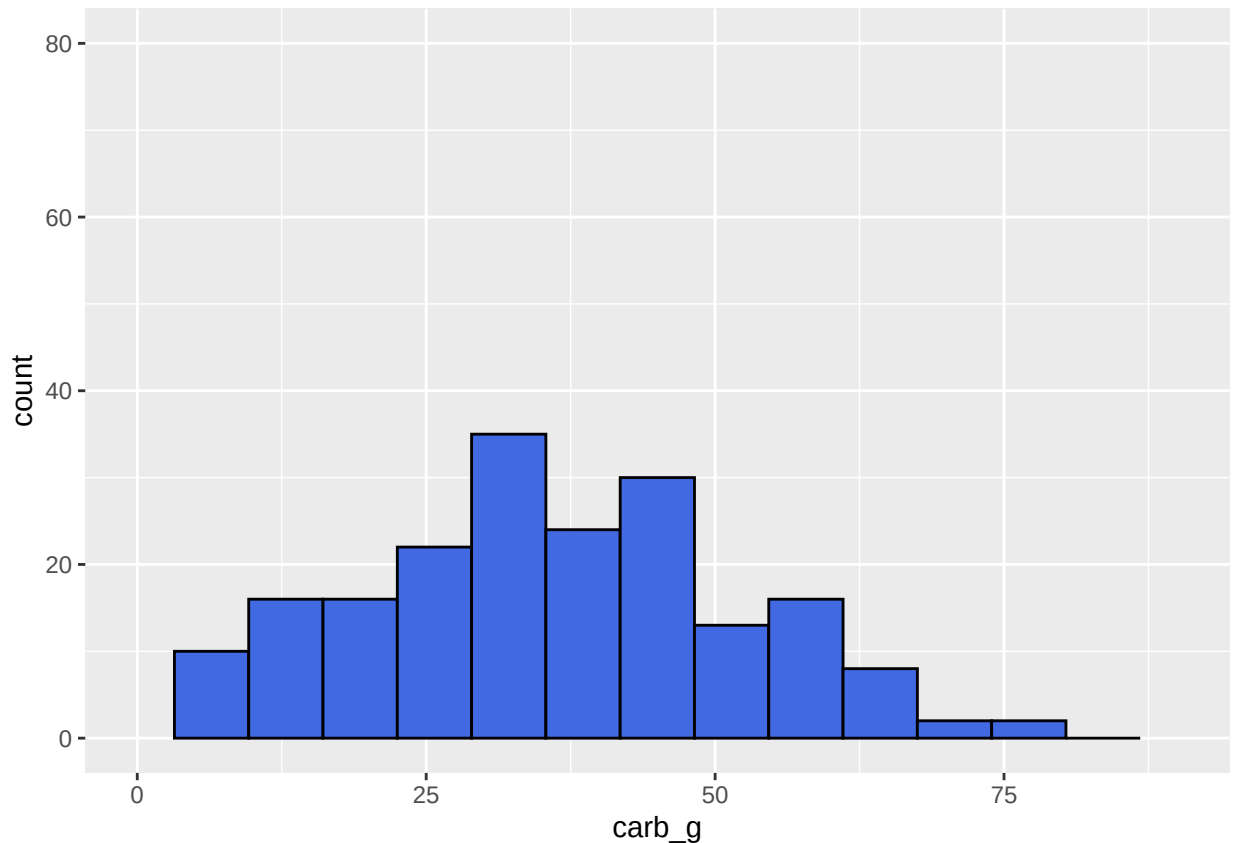
```
star_items_final |>
  ggplot(aes(x = fat_g)) +
  geom_histogram(bins = 15, color = "black", fill = "royalblue") +
  # restrict x-axis for cleaner comparison
  xlim(0, 90) +
  ylim(0, 80)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



```
# For carbs
star_items_final |>
  ggplot(aes(x = carb_g)) +
  geom_histogram(bins = 15, color = "black", fill = "royalblue") +
  xlim(0, 90) +
  ylim(0, 80)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



(We'll discuss ways to polish plots later on. These work for now.)

Fats appear to be far more right-skewed (i.e., more values) that are lower) with lower variability compared to carbs, which appears more normally distributed and more varied.

It would also be worthwhile to compare based on the type of item they are (drink vs. food).

We didn't discuss it yet, but using the `psych::describeBy()` function would accomplish this. (I would encourage you to look at the help documentation for this function.)

```
star_items_final |>
  select(-type) |>
  psych::describeBy(group = star_items_final$type)
```

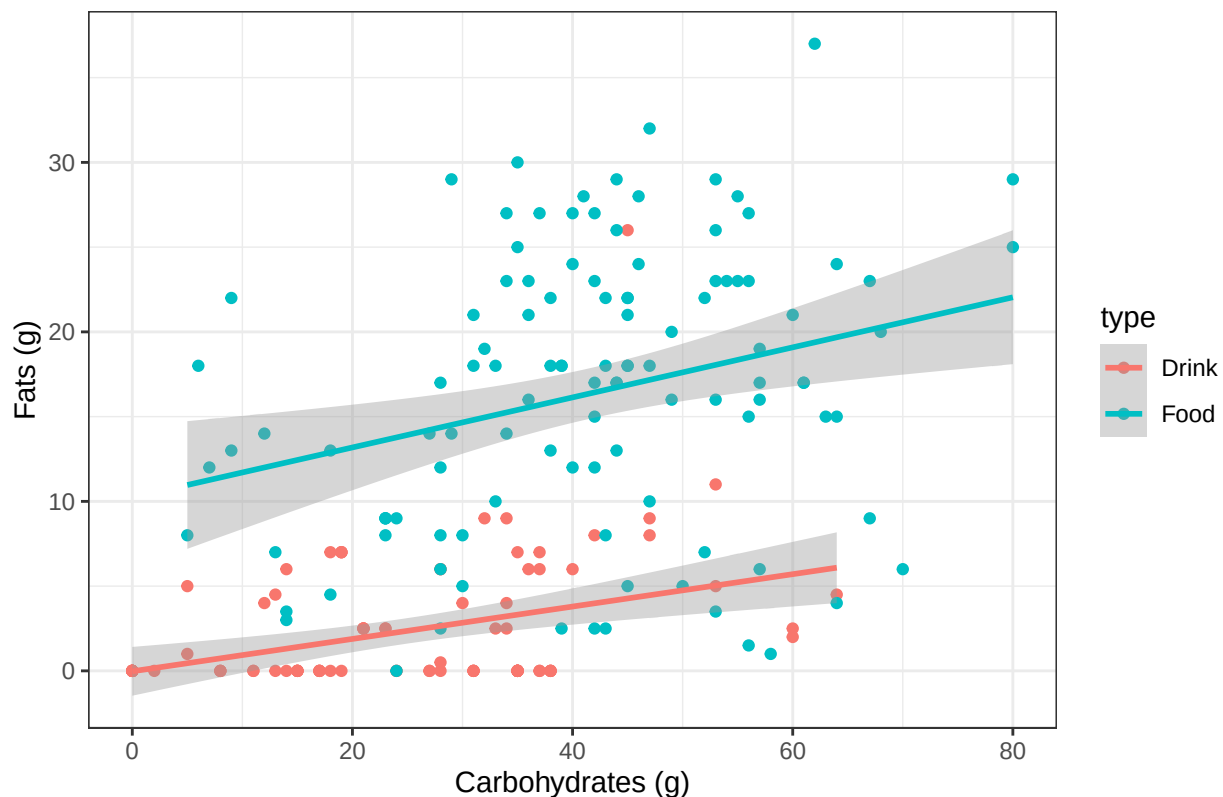
```
##
## Descriptive statistics by group
## group: Drink
##      vars  n  mean    sd median trimmed  mad min max range skew kurtosis
## fat_g    1  92  2.34  3.89    0.0   1.64  0.00  0  26   26  2.85   12.83
## carb_g   2  92 24.74 15.21   25.5  24.39 15.57  0  64   64  0.16   -0.42
##      se
## fat_g  0.41
## carb_g 1.59
## -----
## group: Food
##      vars  n  mean    sd median trimmed  mad min max range skew kurtosis
## fat_g    1 113 16.35  8.3    17   16.43  8.90  0  37   37 -0.08   -0.84
```

```
## carb_g    2 113 41.49 15.8    42  41.91 16.31    5 80    75 -0.16    -0.18
##          se
## fat_g    0.78
## carb_g    1.49
```

Drinks appear to have fewer grams of fat ($M = 2.34$, $SD = 3.89$) and carbohydrates ($M = 27.74$, $SD = 15.21$) content relative to foods (fat: $M = 16.35$, $SD = 8.30$; carbs: $M = 41.49$, $SD = 15.80$).

```
# Main visualization
star_items_final |>
  ggplot(aes(x = carb_g, y = fat_g, color = type)) +
  geom_point() +
  # Good addition: Fits a line of best fit with margin of error
  geom_smooth(method = "lm", formula = "y ~ x") +
  # Good addition: Adds axis labels
  labs(x = "Carbohydrates (g)", y = "Fats (g)",
       title = "Figure 1") +
  # Good addition: Makes theme simpler
  theme_bw()
```

Figure 1



SAMPLE WRITEUP: **DISCLAIMER—PLEASE READ!**

Writing a proper statistical report is not easy. Students are usually tempted to copy-paste any template provided and sub in words, tweak interpretations, and be done. Doing that, while ok in the short term, does not actually help your statistical skills because interpreting data becomes a memorization game.

Rather than relying on a template, do your best to interpret descriptive stats and plots on your own first. After giving it a try, you may refer to a template just to demonstrate a good example to yourself. Always think critically: what aspects of the template work well, and which don't?

If you really want to use a template, please do not simply copy-paste the one below for your own. Doing so is a bad idea because I have my own style of writing and explaining myself. If you wish to make use of it, you should always modify it to suit the data, descriptive stats, and visualizations that are produced. Most importantly, you must write it in your own voice and style.

END OF DISCLAIMER

START WRITEUP

The data set at present represents nutritional information for various items at Starbucks stores. I was interested in determining if there was a linear relationship between the grams of carbohydrates (sugar) and the grams of fat in each item, and whether this relationship differed as a function of the type of Starbucks item (food or drink).

To accomplish this goal, I generated and interpreted simple univariate descriptive statistics and visualizations.

On average, *Starbucks* items tended to contain more grams of sugar ($M = 33.97$, $SD = 17.61$, $Mdn = 35$) than grams of fat ($M = 10.06$, $SD = 9.96$, $Mdn = 7.0$).

As well, drinks appear to have fewer grams of fat ($M = 2.34$, $SD = 3.89$) and carbohydrates ($M = 27.74$, $SD = 15.21$) relative to foods (fat: $M = 16.35$, $SD = 8.30$; carbs: $M = 41.49$, $SD = 15.80$). (This makes sense because drinks often have lower contents of sugars, fats, etc. . . than foods.)

```
star_items_final |>
  ggplot(aes(x = carb_g, y = fat_g, color = type)) +
  geom_point() +
  # Good addition: Fits a line of best fit with margin of error
  geom_smooth(method = "lm", formula = "y ~ x") +
  # Good addition: Adds axis labels
  labs(x = "Carbohydrates (g)", y = "Fats (g)",
       title = "Figure 1") +
  # Good addition: Makes theme simpler
  theme_bw()
```

Figure 1

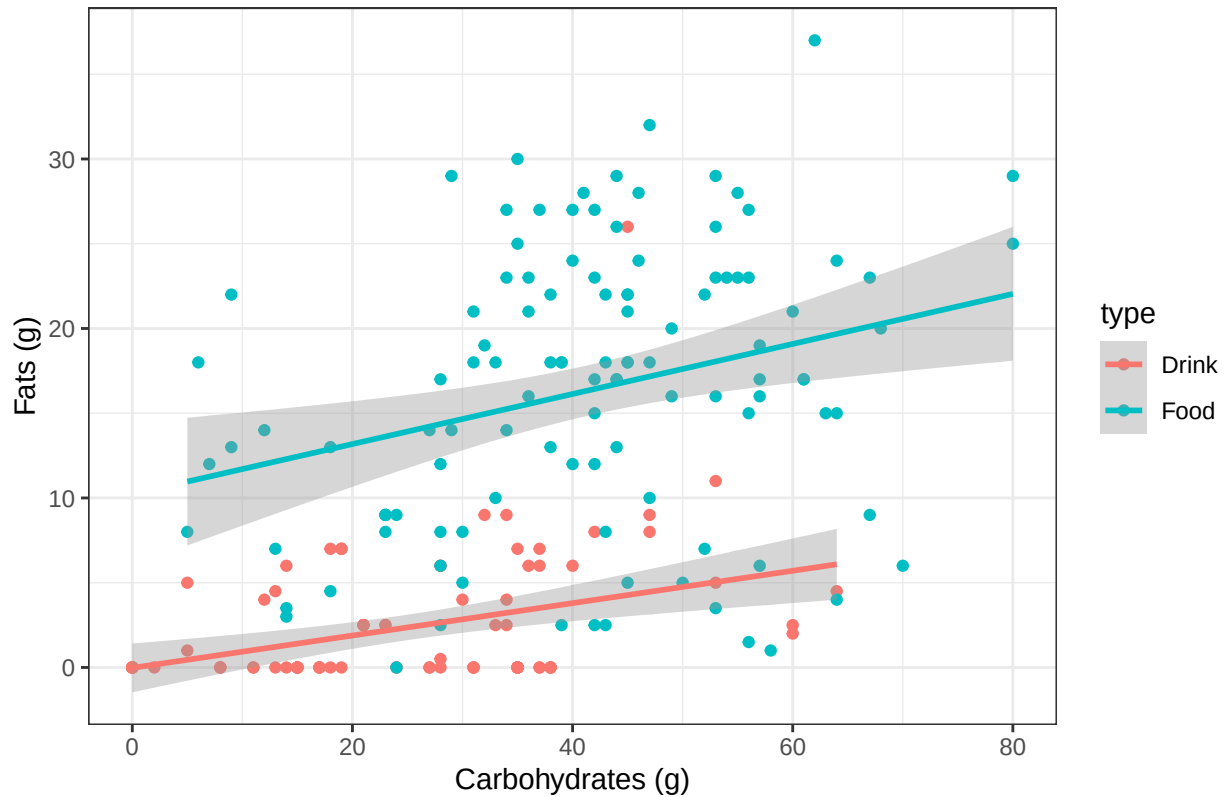


Figure 1 (see above) illustrates the linear relationship between fats and sugar as a function of drink type. In the plot, there is a positive relationship between carbohydrates (g) and fats (g) for both drink and food items. Both relationships tended to have roughly the same linear slope, but had different intercepts.

Overall, this suggests that foods tended to have a higher fat and carbohydrate content relative to drinks, but for foods and drinks, items containing more sugar also tended to contain more fat (and vice versa). Further statistical models should be fit to confirm this relationship and explore it further.

END WRITEUP